

Methodology

1.1. Multilayer MFA

1.1.1. Conceptual framework

TC in simple MFA systems

In Material Flow Analysis (MFAs), Transfer Coefficients (TCs) are a key concept used to quantitatively describe how materials flow through various stages of a system [1–3]. They represent the ratio of material transferred from one stage to another and are instrumental in identifying inefficiencies and bottlenecks in material handling and processing. Additionally, TCs are valuable for scenario modeling and impact prediction. They help evaluate the effects of changes such as the introduction of new technologies, modifications in operational practices, or adjustments to regulatory requirements. Figure 1.1 illustrates a simple MFA system with the mass transfer equations expressed below.



Figure 1.1. A simple mass transfer

$$\begin{cases} m_3 = a_{31}m_1 + a_{32}m_2 \\ m_4 = a_{41}m_1 + a_{42}m_2 \end{cases} \Rightarrow \begin{cases} -a_{31}m_1 - a_{32}m_2 + m_3 = 0 \\ -a_{41}m_1 - a_{42}m_2 + m_4 = 0 \end{cases}$$

Assuming that m_1 and m_2 are known (e.g., equal to y_1 and y_2), we get:

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -a_{31} & -a_{32} & 1 & 0 \\ -a_{41} & -a_{42} & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} m_1 \\ m_2 \\ m_3 \\ m_4 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ 0 \\ 0 \end{bmatrix}$$

Extending TC to multi-layer MFA systems

Next, we examine a single-layer system where the layer accounts for all the chemical elements present in the flows. Figure 1.2 illustrates a system composed of two elements (E_1 in blue, and E_2 in yellow). The corresponding algebraic expression is given below assuming that m_1 to m_4 are known (equal to y_1 to y_4).

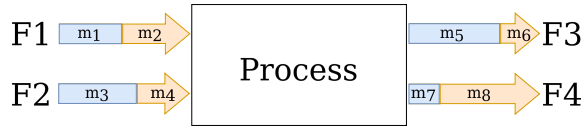


Figure 1.2. A single layer mass transfer where each flow is composed of 2 elements

$$\begin{cases} m_5 = a_{51}m_1 + a_{53}m_3 \\ m_6 = a_{62}m_2 + a_{64}m_4 \\ m_7 = a_{71}m_1 + a_{73}m_3 \\ m_8 = a_{82}m_2 + a_{84}m_4 \end{cases} \Rightarrow \begin{cases} -a_{51}m_1 - a_{53}m_3 + m_5 = 0 \\ -a_{62}m_2 - a_{64}m_4 + m_6 = 0 \\ -a_{71}m_1 - a_{73}m_3 + m_7 = 0 \\ -a_{82}m_2 - a_{84}m_4 + m_8 = 0 \end{cases}$$

$$\begin{bmatrix} 1 & & & & & & & \\ & 1 & & & & & & \\ & & 1 & & & & & \\ & & & 1 & & & & \\ & & & & 1 & & & \\ -a_{51} & & & -a_{53} & & & & \\ & -a_{62} & & & -a_{64} & & & \\ -a_{71} & & & -a_{73} & & & & \\ & -a_{82} & & & -a_{84} & & & \end{bmatrix} \cdot \begin{bmatrix} m_1 \\ m_2 \\ m_3 \\ m_4 \\ m_5 \\ m_6 \\ m_7 \\ m_8 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

This can also be written as:

$$(\mathbf{I} - \mathbf{A})\mathbf{x} = \mathbf{y} \quad (1.1)$$

with $\mathbf{I}$ the identity matrix, $\mathbf{A}$ the technology matrix that records the TCs a_{ij} associated with the mass transfer from m_j to m_i , $\mathbf{x}$ the masses to be determined (i.e., the vector composed of each mass m_i), and $\mathbf{y}$ the inputs. The computational structure is thus similar to LCA and input-output analysis [4–6]. The parallel with the technology matrix from IO tables is further illustrated in figure 1.3.

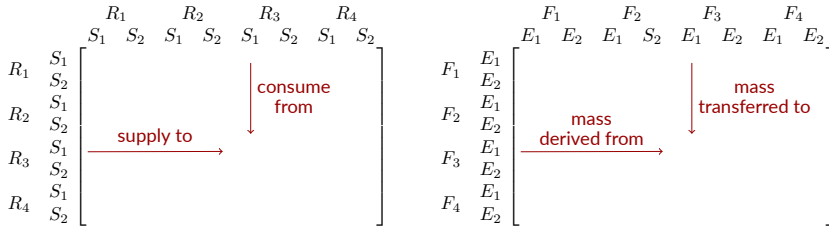


Figure 1.3. (left) example of a technology matrix with 4 regions (R_1, R_2, R_3 and R_4) and 2 sectors (S_1 and S_2) as conceptualised in MRIO tables; (right) technology matrix associated with figure 1.2.

Equation (1.1) can be expanded to encompass any number of layers without loss of generality. Similarly to Multi-Regional Input-Output tables, this is done by considering the combination (i.e., the Cartesian product) of every possible option for each layer. In this study, we focus on four layers: products, components, materials, and elements (figure 1.4).

1.1.2. Implementation

In practice, TCs can be understood as the product of process efficiency and market share. The market shares are directly derived from the previous literature review (??) [7], with each plant assumed to process a portion of the EOL volume based on its capacity. It is further assumed that all batteries are collected, in line with the battery directive and consistent with practices for lead-acid batteries used in internal combustion engine vehicles. The following section details the data collection process for determining the process efficiencies. For a full overview of the model algo-

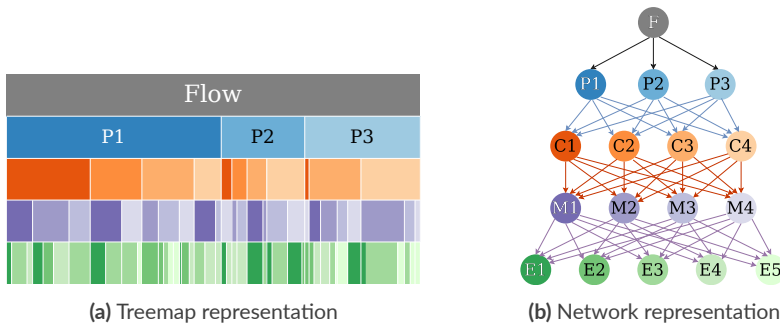


Figure 1.4. Multi-layer representation of a flow as a treemap (left) and a network (right). A network edge $u \rightarrow v$ represents the mass of v contained in u . For clarity, not every edge is represented. For example, there are actually 3 edges between $C1$ and $M1$ depending on if $M1$ is part of $P1$, $P2$ or $P3$.

rithm and its integration with the dynamic MFA (dMFA), we redirect the reader to appendix A.

Deactivation and dismantling

Quantifying the number of recycling plants that rely primarily on manual disassembly is challenging, but conducting the dMFA at the cell level helps mitigate this uncertainty. By focusing on the cell level, the analysis bypasses the need to account for pack and module components such as casings, wires, and coolant which are more likely to be targeted by manual dismantling. Additionally, mechanical pre-treatment can be seen as a form of rough and automated dismantling. Modeling deactivation is equally difficult since the boundary between deactivation and mechanical pre-treatment can be blurry (e.g., see wet crushing). However, from a mass balance perspective, deactivation mainly results in the loss of electrolytes, a loss that also occurs during mechanical treatment. Consequently, the deactivation and dismantling steps are not explicitly modeled.

References

1. Brunner, P. H. & Rechberger, H. *Handbook of Material Flow Analysis: For Environmental, Resource, and Waste Engineers* Second Edition. ISBN: 978-1-4987-2134-9 (CRC Press, Taylor & Francis Group, Boca Raton, 2017).
2. Romagnoli, J. A. & Sánchez, M. C. *Data Processing and Reconciliation for Chemical Process Operations Process Systems Engineering* v. 2. ISBN: 978-0-12-594460-1 (Academic Press, San Diego, 2000).
3. Laner, D. & Rechberger, H. in *Special Types of Life Cycle Assessment* (ed Finkbeiner, M.) 293–332 (Springer Netherlands, Dordrecht, 2016). ISBN: 978-94-017-7610-3. https://doi.org/10.1007/978-94-017-7610-3_7.
4. Heijungs, R., Yang, Y. & Park, H.-S. A or I-A? Unifying the Computational Structures of Process- and IO-based LCA for Clarity and Consistency. *Journal of Industrial Ecology* **26**, 1824–1836. ISSN: 1530-9290. <https://onlinelibrary.wiley.com/doi/abs/10.1111/jiec.13323> (2022).
5. Miller, R. E. & Blair, P. D. *Input-Output Analysis: Foundations and Extensions* 3rd ed. ISBN: 978-1-108-67621-2. <https://www.cambridge.org/core/product/identifier/9781108676212/type/book> (Cambridge University Press, 2021).
6. Heijungs, R. & Suh, S. *The Computational Structure of Life Cycle Assessment* (ed Tukker, A.) ISBN: 978-94-015-9900-9. <https://www.semanticscholar.org/paper/4b80bf5f6254d25f1e0186f54aadf233798668b2> (Springer Netherlands, Dordrecht, 2002).
7. Perello-Y-Bestard, A. *European Lithium Battery Recycling Facilities 2010-2030 2024*. <https://zenodo.org/records/13766892>.
8. Gleiß, A. et al. Identification of Material Flow Systems. *Environmental Science and Pollution Research* **5**, 245–258. ISSN: 1614-7499. <https://doi.org/10.1007/BF02986408> (1998).
9. Bauer, G., Deistler, M., Gleiß, A., Glenck, E. & Matyus, T. Identification of Material Flow Systems. *Environmental Science and Pollution Research* **4**, 105–112. ISSN: 1614-7499. <https://doi.org/10.1007/BF02986290> (1997).
10. Dubois, D., Fargier, H., Ababou, M. & Guyonnet, D. A Fuzzy Constraint-Based Approach to Data Reconciliation in Material Flow Analysis. *International Journal of General Systems* **43**, 787–809. ISSN: 0308-1079. <https://doi.org/10.1080/03081079.2014.920840> (2014).
11. Cencic, O. & Fröhwrth, R. A General Framework for Data Reconciliation—Part I: Linear Constraints. *Computers & Chemical Engineering* **75**, 196–208. ISSN: 0098-1354. <https://www.sciencedirect.com/science/article/pii/S0098135414003330> (2015).
12. Cencic, O. Nonlinear Data Reconciliation in Material Flow Analysis with Software STAN. *Sustainable Environment Research* **26**, 291–298. ISSN: 2468-2039. <https://www.sciencedirect.com/science/article/pii/S246820391630053X> (2016).
13. Kopec, G. M., Allwood, J. M., Cullen, J. M. & Ralph, D. A General Nonlinear Least Squares Data Reconciliation and Estimation Method for Material Flow Analysis. *Journal of Industrial Ecology* **20**, 1038–1049. ISSN: 1530-9290. <https://onlinelibrary.wiley.com/doi/abs/10.1111/jiec.12344> (2016).
14. Cencic, O. & Fröhwrth, R. Data Reconciliation of Nonnormal Observations with Nonlinear Constraints. *Journal of Applied Statistics* **45**, 2411–2428. ISSN: 0266-4763. <https://doi.org/10.1080/02664763.2017.1421916> (2018).
15. Džubur, N., Sunanta, O. & Laner, D. A Fuzzy Set-Based Approach to Data Reconciliation in Material Flow Modeling. *Applied Mathematical Modelling* **43**, 464–480. ISSN: 0307-904X. <https://www.sciencedirect.com/science/article/pii/S0307904X16306254> (2017).
16. Tan, R. R. in *Reference Module in Earth Systems and Environmental Sciences* (Elsevier, 2023). ISBN: 978-0-12-409548-9. <https://www.sciencedirect.com/science/article/pii/B9780323903868000711>.
17. Tan, R. R., Aviso, K. B. & Lee, J.-Y. in *Optimization for Energy Systems and Supply Chains* 1st ed., 151–167 (CRC Press, Boca Raton, 2022). ISBN: 978-1-00-324022-8. <https://www.taylorfrancis.com/books/9781003240228/chapters/10.1201/9781003240228-11>.

18. Duchin, F. & Levine, S. H. Sectors May Use Multiple Technologies Simultaneously: The Rectangular Choice-of-Technology Model with Binding Factor Constraints. *Economic Systems Research* **23**, 281–302. ISSN: 0953-5314. <https://doi.org/10.1080/09535314.2011.571238> (2011).
19. Duchin, F. & Levine, S. H. Choosing among Alternative Technologies: Conditions for Assuring the Feasibility of an Input–Output Database or Scenario. *Economic Systems Research* **29**, 541–556. ISSN: 0953-5314. <https://doi.org/10.1080/09535314.2017.1301396> (2017).
20. Scipy. Solve — SciPy v1.14.0 Manual <https://docs.scipy.org/doc/scipy/reference/generated/scipy.linalg.solve.html> (07/23/2024).
21. Wikipedia. Sparse Matrix https://www.wikiwand.com/en/Sparse_matrix (07/23/2024).
22. Scipy. Spsolve — SciPy v1.14.0 Manual <https://docs.scipy.org/doc/scipy/reference/generated/scipy.sparse.linalg.spsolve.html> (07/23/2024).

Appendix

Multilayer MFA

A.1. System determinism

In the context of this work, we assume that the system is both consistent and uniquely determined, meaning it has exactly one solution, and we can assume the technological matrix A to be square. Various optimization techniques have been proposed to handle situations where there are either more equations than unknowns (over-determined systems) [2, 8–15] or more unknowns than equations (under-determined systems) [16–19]. However, adapting these methods to multi-layer MFA is beyond the scope of this study.

A.2. Model interface

To solve the system's equation (1.1), specifying initial conditions in addition to the Transfer Coefficients (TCs) is necessary. These initial conditions include (i) the total amount of products collected for waste processing and (ii) the mass composition of those products. Excel files are the simplest format for providing these inputs, and they can be easily structured to map to the technological matrix A . Excel tables can also serve as an interface for the framework and enable users with limited programming or mathematical background to develop their own models. Consequently, the format of the tables should be generic and straightforward enough to not hinder the data collection process, while also being adaptable to the specific compositional structures of the system under study.

The figures A.1 to A.3 illustrate the format that was selected. To simplify the data collection process, the formats are different for each dataset. The inflow composition explicitly requires the user to specify each layer (product, component, material, element). In contrast, the data format for

mass inflow (figure A.1) only requires the upper level (typically the product layer), and the data format for TCs (figure A.3) only needs the flow IDs and a single layer. This approach aims to simplify the data collection process.

Flow ID	Layer 1	Mass (kt)
F_{in}	P_1	2.5
F_{in}	P_2	1.7

Figure A.1. Data format for mass inflow.

Flow ID	Layer 1	Layer 2	Layer 3	Layer 4	Share
F_{in}	P_1	C_1			0.4
F_{in}	P_1	C_1	M_1		0.3
F_{in}	P_1	C_1	M_1	E_1	0.25

Figure A.2. Data format for inflow composition. The first row indicates that component C_1 makes up 40% of the mass of product P_1 . The second row shows that material M_1 represents 30% of the mass of component C_1 , which is embedded in product P_1 . Similarly, the last row states that element E_1 represents 25% of the mass of material M_1 within component C_1 in product P_1 .

Process	Inflow		Outflow		TC
T_1	F_6	P_1	F_3	P_1	0.5
T_1	F_6	M_2	F_3	E_3	0.7
T_3	F_5	E_1	F_8	E_1	0.9

Figure A.3. Data format for the TCs. The first row indicates that 50% of product P_1 is recovered through process T_1 . The second row illustrates a disassembly process where 70% of the element E_3 embedded in M_2 is recovered. The last row means that 90% of E_1 present in F_5 is recovered via process T_3 , regardless of what it was embedded in.

A.3. Algorithm

The algorithm consists of three steps. The first one is to restructure the Excel tables to ensure every layer is represented. This restructuring is straightforward for the mass inflow data, which is used to define the y vector in the equation $(\mathbf{I}-\mathbf{A})\mathbf{x} = \mathbf{y}$ (figure A.4), and for the TCs, where the corresponding rows and columns in the technology matrix can be easily derived (figure A.6). For the composition data, it can be noted that mathematically, expressing the mass of a sublayer (e.g., a component) based on the layer in which it is embedded (e.g., the product) and the corresponding composition fraction is equivalent to using the previously discussed

TC formulation. This means that the technology matrix **A** can also incorporate these composition fractions (figure A.5).

Flow ID	Layer 1	Mass (kt)
F_{in}	P_1	2.5
F_{in}	P_2	1.7

⇓

Row index					Mass (kt)
F_{in}	P_1	$\emptyset$	$\emptyset$	$\emptyset$	2.5
F_{in}	P_1	$\emptyset$	$\emptyset$	$\emptyset$	1.7

Figure A.4. Restructuring the Excel data on mass inflows to ensure all layers are represented. Row indexes refers to the indexes of the **y** vector.

Flow ID	Layer 1	Layer 2	Layer 3	Layer 4	Share
F_{in}	P_1	C_1			0.4
F_{in}	P_1	C_1	M_1		0.3
F_{in}	P_1	C_1	M_1	E_1	0.25

⇓

Column index					Row index					share
F_{in}	P_1	$\emptyset$	$\emptyset$	$\emptyset$	F_{in}	P_1	C_1	$\emptyset$	$\emptyset$	0.4
F_{in}	P_1	C_1	$\emptyset$	$\emptyset$	F_{in}	P_1	C_1	M_1	$\emptyset$	0.3
F_{in}	P_1	C_1	M_1	$\emptyset$	F_{in}	P_1	C_1	M_1	E_1	0.2

Figure A.5. Restructuring Excel data on compositions. Restructuring follows from the observation that $\text{mass}([F_{in}, P_1, C_1]) = 0.4 \cdot \text{mass}([F_{in}, P_1])$ and similarly for other rows. Row and column indexes refer to the indexes of the **A** matrix.

Inflow		Outflow		TC
F_4	C_2	F_7	M_1	0.18

⇓

Column index					Row index					TC
F_4		C_2			F_7			M_1		0.18

⇓

Column index					Row index					TC
F_4		C_2	M_1		F_7		C_2	M_1		0.18

Figure A.6. Restructuring Excel data on TCs to ensure all layers are represented, with placeholders maintained in anticipation of step 2. Row and column indexes refer to the indexes of the **A** matrix.

The algorithm's second step involves inferring all possible combinations of products, components, materials, and elements that a row from the TC table should apply to. The underlying assumption is that users will specify TCs for the most relevant layer rather than for every possible combination of embedded sub-layers. For instance, if a component has a recovery rate of 50%, it is assumed that all materials and elements within that component are recovered at the same rate. Similarly, users might specify a default recovery value for a chemical element without knowing exactly which component or product it will be part of when it reaches this particular process. To accommodate such cases, the empty symbol is used, ensuring that every combination of flow, product, component, material, and element can be represented as an ordered sequence of length 5, even when dealing with aggregated entities or when an upper layer becomes irrelevant (see figure A.7). This inference process is achieved by generating the Cartesian product, as shown in figure A.8. Unlike for TCs, this step is not required for composition data, as it is assumed that users will provide complete composition information for the system, which cannot be inferred otherwise.

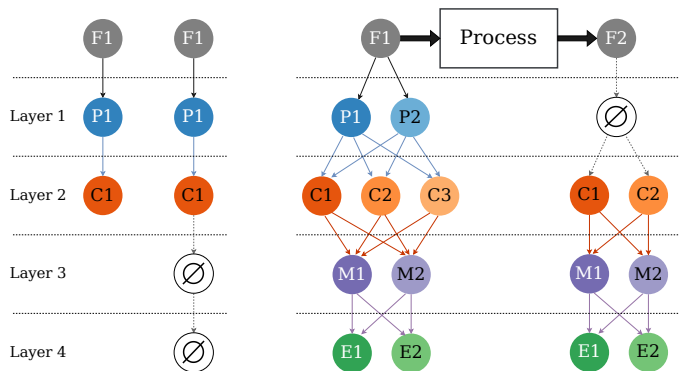


Figure A.7. The symbol $\emptyset$ is used to maintain a fixed length ordered sequence (F, P, C, M, E) when considering an aggregated entity (left) or when an upper level is no longer present (right).

The third step involves populating the vector y and the technology matrix A with the newly obtained data. However, for the technology matrix, this process may introduce potential conflicts that could produce incorrect outputs and be difficult to detect. Indeed, the expansion from the second step can lead to a sequence being assigned multiple TCs. For ex-

Column index					Row index					TC
F_4		C_2	M_1		F_7		C_2	M_1		0.18

⇓

Column index					Row index					TC
F_4	P_1	C_2	M_1	E_1	F_7	P_1	C_2	M_1	E_1	0.18
F_4	P_1	C_2	M_1	E_2	F_7	P_1	C_2	M_1	E_2	0.18
F_4	P_1	C_2	M_1	E_3	F_7	P_1	C_2	M_1	E_3	0.18
	...					...				...
F_4	P_2	C_2	M_1	E_1	F_7	P_2	C_2	M_1	E_1	0.18
F_4	P_2	C_2	M_1	E_2	F_7	P_2	C_2	M_1	E_2	0.18
F_4	P_2	C_2	M_1	E_3	F_7	P_2	C_2	M_1	E_3	0.18
	...					...				...
F_4	$\emptyset$	C_2	M_1	E_1	F_7	$\emptyset$	C_2	M_1	E_1	0.18
F_4	$\emptyset$	C_2	M_1	E_2	F_7	$\emptyset$	C_2	M_1	E_2	0.18
F_4	$\emptyset$	C_2	M_1	E_3	F_7	$\emptyset$	C_2	M_1	E_3	0.18
	...					...				...

Figure A.8. The second step of the algorithm implementation is to infer the TC for the missing combinations of products, components, materials, and elements.

ample if for the same process the user specify a TC α for a component C_1 , and at the same time assume a TC β for material M_1 , then the sequence $[C_1, M_1]$ will be associated to both α and β . This is incompatible with the assumption of a determined system, as only a single TC should be used to ensure that the technology matrix $\mathbf{A}$ is square. By default, TCs specified closer to the chemical element are considered more reliable and are used to populate the matrix. However, any conflicts can be visualised to allow the user to review and verify the matrix entries. It would also be trivial to add a column to the TCs Excel file to specify the priority (i.e., the certainty) of each coefficient

Figure A.9 illustrates the integration of mass composition and TCs within the technology matrix $\mathbf{A}$ following the three steps described above. Finally, the equation $(\mathbf{I} - \mathbf{A})\mathbf{x} = \mathbf{y}$ can be solved with a simple linear algebra solver such as the one provided by the Python's Scipy library [20].

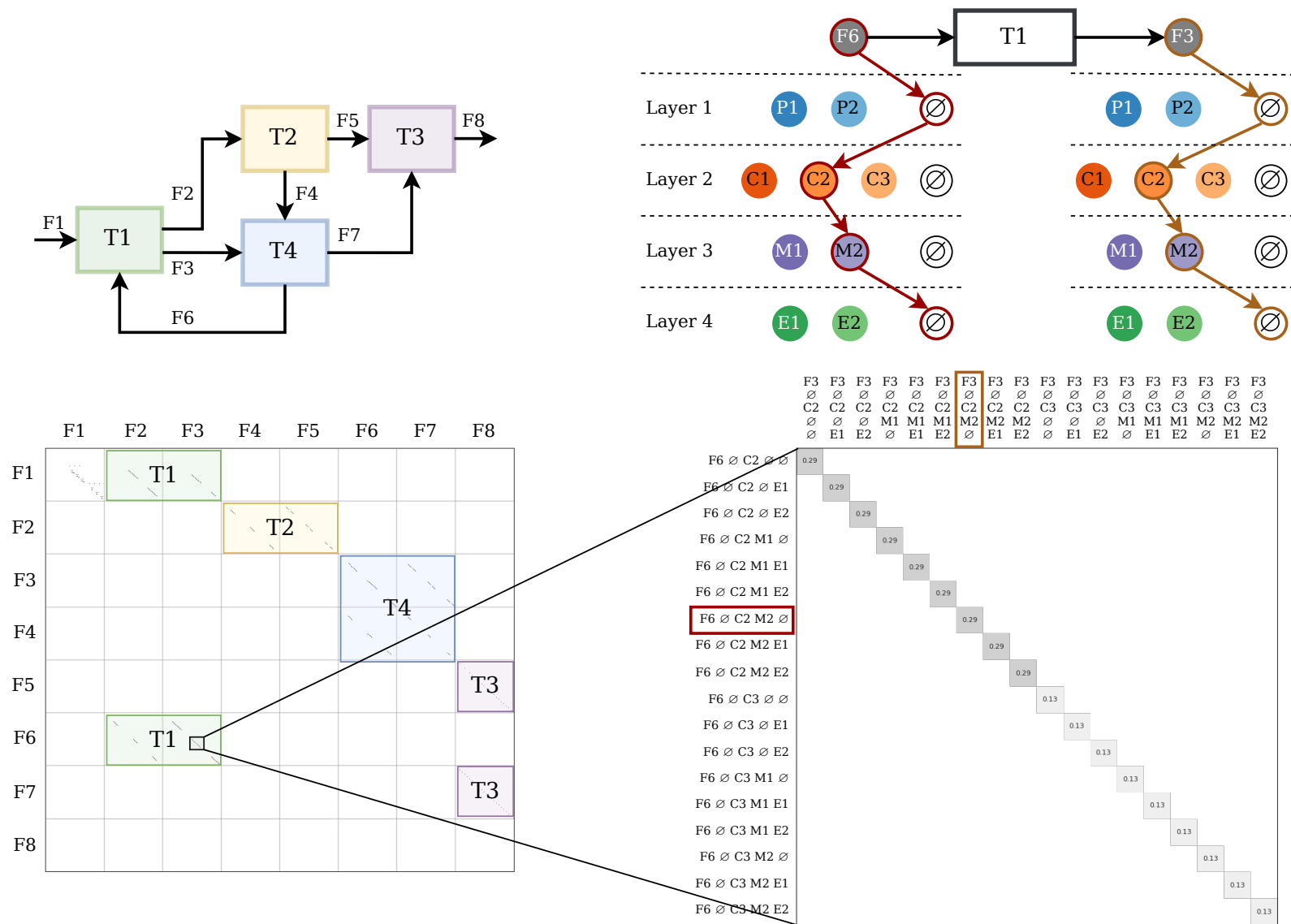


Figure A.9. A multi-layer MFA system (top) and its transposed technology matrix (bottom). The cell $(F1, F1)$ in the technology matrix represents the mass composition of the inflow $F1$.

Table A.1. Flow equations derived from the system depicted in figure A.9. By contention, inflows are positive and outflows are negative. The table can then be used to verify the mass balance of the system.

Process	F_1	F_2	F_3	F_4	F_5	F_6	F_7	F_8
T_1	1	-1	-1			1		
T_2		1		-1	-1			
T_3					1		1	-1
T_4			1	1		-1	-1	

A.4. Model output and mass balance

The output of the static multi-layer MFA reproduces the same format as the inflow composition (figure A.1). Additionally, a table with the flow equations is generated along with the solution (see table A.1 for more details). This table is then used to verify the mass balance of the system. However, strict enforcement of mass balance is not applied for three reasons: i) mass balance only applies at the chemical element layer, therefore attention to edge cases should be paid when applying mass balance to more aggregated levels (e.g., a product no longer exist after disassembly, and only the components remain); ii) some chemical elements may not be of interest to the modeler, in which case TCs might not have been provided (e.g. the amount of carbon or oxygen released by air emissions during incineration); iii) uncertainty and sensitivity analysis may be conducted to identify hotspots, and the value ranges may not necessarily add up to 100%.

A.5. Computational complexity

The algorithm's simplicity and maintainability come with a trade-off: exponential complexity in both space and time. As shown in figure A.9, most of the technology matrix is empty, making it inefficient to store it as a full square matrix. To address space complexity, the matrix is stored as a Compressed Sparse Row (CSR) matrix [21] and equation (1.1) is processed using SciPy's wrapper of the UMFPACK sparse solver [22]. Converting the square technology matrix $\mathbf{A}$ into a CSR format requires first transforming the ordered sequence $[F_m, P_n, C_o, M_p, E_q]$, which is used for indexing and retrieving TCs within the technology matrix, into a unique integer. To

achieve this, we assign a unique integer to each product, component, material, and element: $[m, n, o, p, q]$. We then define a bijective mapping to establish a correspondence between such a sequence and a unique integer. Let $\mathbf{s}$ be the vector representing the number of variables for each layer, with d layers:

$$\mathbf{s} = [s_0, s_1, \dots, s_{d-1}]$$

and the linear indexing coefficients $\mathbf{c} = [c_0, c_1, \dots, c_{d-1}]$, where each coefficient c_i is given by:

$$c_i = \begin{cases} \prod_{j=i+1}^{d-1} s_j & \text{for } i < d-1 \\ 1 & \text{for } i = d-1 \end{cases}$$

Let $\alpha = [\alpha_0, \alpha_1, \dots, \alpha_{d-1}]$ be the vector that represent the sequence $[F_m, P_n, C_o, M_p, E_q]$. Its unique scalar index k is computed as:

$$k = \sum_{i=0}^{d-1} \alpha_i \cdot c_i$$

$$\begin{bmatrix} F_1 \\ F_2 \\ F_3 \\ F_4 \\ F_5 \\ F_6 \end{bmatrix} \begin{bmatrix} P_1 \\ P_2 \\ P_3 \\ P_4 \\ \emptyset \end{bmatrix} \begin{bmatrix} C_1 \\ C_2 \\ C_3 \\ C_4 \\ C_5 \\ C_6 \\ C_7 \\ C_8 \\ \emptyset \end{bmatrix} \begin{bmatrix} M_1 \\ M_2 \\ M_3 \\ M_4 \\ M_5 \\ M_6 \\ \emptyset \end{bmatrix} \begin{bmatrix} E_1 \\ E_2 \\ E_3 \\ E_4 \\ E_5 \\ E_6 \\ E_7 \\ E_8 \\ E_9 \\ \emptyset \end{bmatrix}$$

Figure A.10. A bijective mapping ensures a correspondence between a sequence and a unique integer. Here, the index of $[F_3, P_1, C_4, M_2, E_8]$ is 10 388. This index is then used to convert the technology matrix $\mathbf{A}$ into a CSR format.

Further optimization is necessary due to the RAM usage of the sparse solver, which often becomes the limiting factor. To mitigate this, two non-exclusive approaches can be employed, though they increase time complexity: i) Matrix Reduction: The size of the technology matrix can be reduced by recognizing that, due to the mass conservation principle, the element layers are independent of each other. This allows for the generation of a smaller technology matrix for each chemical element. The so-

lution for each sub-system can then be computed independently and aggregated afterward. ii) Graph Partitioning: An MFA system can be viewed as a graph network that may contain cycles. By identifying these cycles and aggregating the nodes within each loop into a "virtual" node (i.e., partitioning the graph into strongly connected components, see figure A.11), the graph can be converted into a "virtual" directed acyclic graph (DAG). Once this DAG is obtained, a topological sort can order these "virtual" nodes, effectively dividing the overall MFA system into sub-systems that can be processed sequentially.

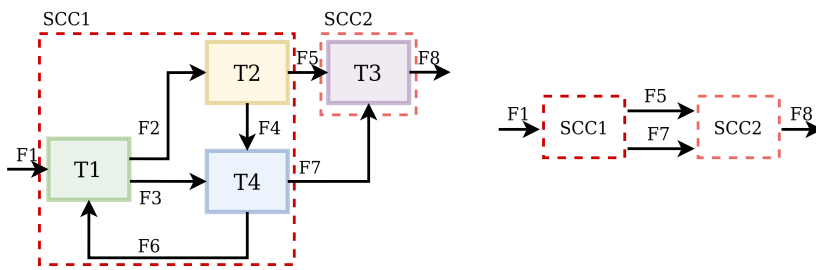


Figure A.11. Example of sub-systems division.